1.输入输出，类的入门

```C++
/*#include<stdio.h>
int main()
{int a,b;
  scanf("%d%d",&a,&b);
  printf("%d",a+b);}
*/
#include<iostream>
using namespace std;
int main()
{double a,b;
  int x=999;
  cin>>a>>b;//输入
  cout<<a+b<<endl;//输出 相加
  cout<<a<<"+"<<b<<"="<<a+b<<' '<<x<<endl;//end of line}
```

```C++
//要求：使用switch语句，使用cout
//输入输出示例：
//输入1：1
//输出1：Monday
//输入2：7
//输出2：Sunday
#include<iostream>
using namespace std;
int main(){
  int day;
  cin>>day;
  switch(day){
    case 1:
      cout<<"Monday"<<endl;
      break;
    case 2:
      cout<<"Tuesday"<<endl;
      break;
    case 3:
      cout<<"Wednesday"<<endl;
      break;
    case 4:
      cout<<"Thursday"<<endl;
      break;
    case 5:
      cout<<"Friday"<<endl;
      break;
    case 6:
      cout<<"Saturday"<<endl;
      break;
    case 7:
      cout<<"Sunday"<<endl;
      break;
    default:
      cout<<"error"<<endl;
      break; }
  return 0;
}
```

```C++
#include <iostream>
using namespace std;
class fractionAdd{
private:
  int numerator;
  int denominator;
  int gcd(int a,int b){
    while(b!=0)
    {int r=a%b;
      a=b;
      b=r;}
    return a;//辗转相除求最大公因数
    }
public:
   fractionAdd(int n1,int n2,int n3,int n4){
```

```
      numerator=n1*n4+n2*n3;
      denominator=n2*n4;
       int divisor=gcd(numerator,denominator);
      numerator/=divisor;
      denominator/=divisor;}
      void Show(){cout<<numerator<<"/"<<denominator<<endl;}
};
int main(){
  int a,b,c,d;
  cin>>a>>b>>c>>d;
  fractionAdd frac(a,b,c,d);
  frac.Show();
  return 0;}
```

```C++
#include<iostream>
using namespace std;
class Fraction{
private:
  int numerator;
  int denominator;
  int gcd(int a,int b)
  {while(b!=0)
  {int r=a%b;
  a=b;
  b=r;
  }
  return a;
  }
public:
  Fraction(int n,int d)
  {
    int divisor=gcd(n,d);
    numerator=n/divisor;
    denominator=d/divisor;
  }
  Fraction(int n)
  {
    numerator=n;
    denominator=1;
  }
  Fraction()
  {
    numerator=1;
    denominator=1;//重载
  }
  void show()
  {
    cout<<numerator<<"/"<<denominator;
  }
};
//StudybarCommentBegin
int main()
{
  Fraction f1,f2(2),f3(1,3);
  f1.show(); cout<<endl;
  f2.show(); cout<<endl;
  f3.show(); cout<<endl;
}
//StudybarCommentEnd
```

时间类

```C++
#include<iostream>//基本时间类
using namespace std;
class Cmytime{
  private:
    int h,m,s;
  public:
    void Set(int hh,int mm,int ss){h=hh;m=mm;s=ss;};
    void Show(){cout<<h<<":"<<m<<":"<<s<<endl;};
};
//StudybarCommentBegin
int main(void) {
    int h,m,s;
    cin>>h>>m>>s;
    Cmytime t1;//t1是对象，是零件，object
  t1.Set(h,m,s);
  t1.Show();
    return 0;
}
//StudybarCommentEnd
```

```cpp
#include<iostream>
using namespace std;
class Cmytime{
  private:
    int h,m,s;
  public:
    Cmytime()
    {}
    Cmytime(int hh,int mm,int ss)
    {
    h=hh;
    m=mm;
    s=ss;
    }
    void Set(int hh,int mm,int ss){h=hh;m=mm;s=ss;};
    void Show(){cout<<h<<":"<<m<<":"<<s<<endl;};
};
//StudybarCommentBegin
int main(void) {
  int h,m,s;
  cin>>h>>m>>s;
  Cmytime t1(3,2,1);
  t1.Show();
  cout<<"\n";
//  Cmytime t3;
//  t3.Show();
//  cout<<"\n";
  t1.Set(h,m,s);
  t1.Show();
  return 0;
}
//StudybarCommentEnd
```

```cpp
#include<iostream>
using namespace std;
class Cmytime{
  private:
    int h,m,s;
  public:
    int Set(int hh,int mm,int ss){
    if (0<=hh&&hh<=23&&0<=mm&&mm<=59&&0<=ss&&ss<=59){
    h=hh;
    m=mm;
    s=ss;
    return 1;}
    else{
  //  cout<<"invalid value"<<endl;
    return 0;}
    }

    void AddOneSecond(){//时间加一秒
    s=s+1;
    if(s>60){
    s=s-60;
    m=m+1;
    }
    if(m>60){
    m=m-60;
    h=h+1;
    }
    if(h>23)
    {
    h=h-24;
    }
    }
    void Show(){
    cout<<h<<":"<<m<<":"<<s<<endl;
    }
};

//StudybarCommentBegin
int main(void) {
    int h,m,s;
   cin>>h>>m>>s;

   Cmytime t1;
   t1.Set(h,m,s);
   t1.Show();
   cout<<endl<<t1.Set(24,0,0)<<"\n";
   t1.Show();

```

```
    t1.AddOneSecond();
    cout<<endl;
    t1.Show();

     return 0;
 }
 //StudybarCommentEnd
```

C++

```
#include<iostream>
using namespace std;
class Cmytime{
  private:
    int h,m,s;
  public:
    int Set(int hh,int mm,int ss){
    if (0<=hh&&hh<=23&&0<=mm&&mm<=59&&0<=ss&&ss<=59){
    h=hh;
    m=mm;
    s=ss;
    return 1;}
    else{
  //  cout<<"invalid value"<<endl;
    return 0;}
    }

   int AddNSeconds(int n) {//时间加N秒
        s = s + m * 60 + h * 3600 + n;
        h = s / 3600;
        m = (s % 3600) / 60;
        s = s % 60;
        int day=h/24;
        return day;
    }
     void Show() {
        cout << h << ":" << m << ":" << s;
    }
};

//StudybarCommentBegin
int main(void) {
    int h,m,s;
   cin>>h>>m>>s;
   Cmytime t1;
   t1.Set(h,m,s);
   t1.Show();
   cout<<endl<<t1.Set(24,0,0)<<"\n";
   t1.Show();
   t1.AddNSeconds(1);
   cout<<endl;
   t1.Show();
   cout<<endl<<t1.AddNSeconds(3600*25);
   return 0;
}//StudybarCommentEnd
```

C++

```
#include<iostream>
#include<iomanip>
#include <cmath>
using namespace std;
class Cmycomplex{
  private:
    double x,y;
  public:
    Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;}
    void Set(double xx,double yy)
    {
    x=xx;y=yy;
    }
    Cmycomplex operator+(const Cmycomplex&z)//返回值类型是Cmycomplex
    {
      Cmycomplex t;
      t.x=x+z.x;
      t.y=y+z.y;
      return t;
    }
    Cmycomplex operator-(const Cmycomplex&z)//返回值类型是Cmycomplex
    {
      Cmycomplex t;
      t.x=x-z.x;
      t.y=y-z.y;
      return t;
    }
    Cmycomplex operator*(const Cmycomplex&z)//返回值类型是Cmycomplex
```

```
    {
      Cmycomplex t;
      t.x=x*z.x-y*z.y;
      t.y=x*z.y+z.x*y;
      return t;
    }
    Cmycomplex operator/(const Cmycomplex&z)//返回值类型是Cmycomplex
    {
      Cmycomplex t;
      t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
      t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
      return t;
    }

    Cmycomplex sqrt()  //返回值类型是Cmycomplex
    {
    if(x<0 && y==0)
    {
      Cmycomplex t;
      t.x=0;
      t.y=::sqrt(-x);
      return t;
    }
    if (y == 0 && x == 0) {
      return Cmycomplex(0, 0);
        }
      Cmycomplex t;
      t.x=::sqrt((x+::sqrt(x*x+y*y))/2);
      t.y=y/::sqrt(2*(x+::sqrt(x*x+y*y)));
      return t;
    }
    void Show(){
    //具体显示几位小数的设置

    std::cout<<"("<<fixed<<setprecision(2)<<x;//声明空间
    if (y>=0){
        std::cout<< "+"<<fixed<<setprecision(2)<<y<< "i)"<<endl;}
    if (y<0){
        std::cout<<fixed<<setprecision(2)<<y<< "i)"<<endl;}
    }
    double gety()const{
      return y;
    }
    friend Cmycomplex operator*(const double&n,const Cmycomplex&z);
    friend Cmycomplex operator/(const double&n,const Cmycomplex&z);//返回值类型是Cmycomplex

  };
    Cmycomplex operator*(const double&n, const Cmycomplex&z){//友元函数
      Cmycomplex t;
      t.x=n*z.x;
      t.y=n*z.y;
      return t;}
    Cmycomplex operator/(const double&n, const Cmycomplex&z){
      Cmycomplex t;
      t.x=z.x/n;
      t.y=z.y/n;
      return t;}

int main()
{
  double x1, x2, x3, y1, y2, y3;
  cin >> x1 >> y1;
  cin >> x2 >> y2;
  cin >> x3 >> y3;
  Cmycomplex a(x1, y1), b(x2, y2), c(x3, y3), z, t1, t2;
  z=b*b-4*a*c;
  t1=((-1)*b+z.sqrt())/(2*a);     //z.sqrt()为求复数z的平方根，这里的2*a涉及操作符重载，且友元重载。
  t2=((-1)*b-z.sqrt())/2.0/a;   //这里涉及到除法的重载
  if(t1.gety()>t2.gety())   //gety()为得到虚部值
  {
    t1.Show();
    t2.Show();
  }
  else
  {
    t2.Show();
    t1.Show();
  }
  return 0;
}
```